Fiscal Year:	FY 2013	**Task Last Updated:**	FY 12/19/2013
PI Name:	Ferkul, Paul Ph.D.		
Project Title:	SPICE-BASS: Smoking Point in Co-Flow Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Glenn Research Center		
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PI Address 2:	National Center for Space Exploration Research		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Combustion Rapid Turn Around Flight Opportunity
Start Date:	10/01/2009	**End Date:**	06/30/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-133 ; ISS NOTE: New end date is 6/30/2013 per PI (Ed., 6/8/2012) NOTE: project to be extended per PI (Ed., 1/19/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olson, Sandra (NASA Glenn Research Center) Takahashi, Fumiaki (National Center for Space Exploration Research) T'ien, James (Case Western Reserve University)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The experiments described here are based on the hypothesis that materials in microgravity, with adequate ventilation, burn as well if not better than the same material in normal gravity with other conditions being identical (pressure, oxygen concentration, temperature, etc.) To test this hypothesis with any but the thinnest materials requires extended microgravity time – longer than available in drop towers or parabolic aircraft. If this hypothesis is true, it implies serious safety concerns over the utility and margin of safety provided NASA-STD-6001 Test #1. There are important differences in the suppression of fires in space compared to on Earth. On Earth it is understood that best results are generally obtained when the extinguisher "attacks" the base of the flame, which is both the stabilization point and the point of fresh air entrainment. For a fire burning in microgravity, the best point of application of suppressant may not be immediately apparent, especially for a partially obstructed flame or a wake-stabilized flame. Depending on the geometry of the flame and the characteristics of the extinguisher (distance from flame, dispersion angle) it is possible that the suppressant stream will be ineffective or might actually make the flame worse through the entrainment of oxygen. There are two main objectives: 1.) Begin to bridge the gap between the normal gravity NASA-STD-6001 Test#1 method, ground-based microgravity tests, and actual material flammability in microgravity. The relation between these can be investigated by observing flames burning for the longer times available in flight. Practical, realistic (thicker) fuels in typical geometries will be examined, including wake flames which have characteristics of a fire burning behind another object (for example, a fire burning in tight area which may not be easily accessible). 2.) Assess the effectiveness of an inert, gaseous extinguishing agent (similar to that used on ISS) in putting out flames over different materials, geometries, preheating, and flow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space Applications: The current NASA spacecraft materials selection is based on a standard test method (NASA–STD–6001 Test 1) that segregates material based on 1-g behavior without consideration of low gravity effects. A critical element of this understanding is the radiative heat emission from the flame. These results are used in first order models and predictions of heat release in spacecraft fires and as a means to extend heat release data from tests like the NASA cone calorimeter test (NASA–STD–6001 Test 2) to a performance-based material selection process. Using nitrogen as a flame suppressant in microgravity provides a direct link to current and planned extinguishment techniques. Earth Applications: BASS results provide essential guidance to ground-based microgravity combustion research efforts. Detailed combustion models are validated using the simpler flow environment afforded by tests in microgravity. Once validated, they can be used to build more complex combustion models needed to capture the important details of flames burning in normal gravity. These models have wide applicability to the general understanding of many terrestrial combustion problems.
Task Progress:	Microgravity combustion tests burning thin and thick flat samples, acrylic spheres, and candles were conducted aboard the International Space Station. The samples were mounted inside a small wind tunnel which could impose air flow speeds up to 40 cm/s. The wind tunnel was installed in the Microgravity Science Glovebox which supplied power, imaging, and a level of containment. The effects of air flow speed, fuel thickness, and fuel preheating on flame appearance, flame growth, and spread rates were determined in both the opposed and concurrent-flow configuration. For thin fuels in the opposed-flow configuration, the flame quickly reached steady spread for each flow speed, and the spread rate was fastest at an intermediate value of flow speed. These tests show the enhanced flammability in microgravity for this geometry, since, in normal gravity air, a flame self-extinguishes in the opposed-flow geometry (downward flame spread). In the concurrent-flow configuration, flame size grew with time during the tests. A limiting length and steady spread rate were obtained only in low flow speeds (= 10 cm/s) for the short-length samples that fit in the small wind tunnel. For these conditions, flame spread rate increased linearly with increasing flow. This is the first time that detailed transient flame growth data was obtained in purely forced flows in microgravity. In addition, by decreasing flow speed to a very low value (around 1 cm/s), quenching extinction was observed. The valuable results from these long-duration experiments validate a number of theoretical predictions and also provide the data for a transient flame growth model under development. For thick fuels, microgravity flames were found to be especially sensitive to air flow speed in the range 0 to 5 cm/s. The gas phase response is much faster compared to the solid and so as the flow speed is changed, the flame responds with almost no delay. At the lowest speeds examined (less than 1 cm/s) all the flames tended to become dim blue and very stable. However, heat loss at these very low convective rates is small so the flames can burn for a long time. At moderate flow speeds (between about 1 and 5 cm/s) the flame continually heats the solid fuel resulting in an increasing fuel temperature, higher rate of fuel vaporization, and a stronger, more luminous flame as time progresses. Only the smallest flames burning acrylic slabs appeared to be adversely influenced by solid conductive heat loss, but even these burned for over 5 minutes before self-extinguishing. This has implications for spacecraft fire safety since a tiny flame might be undetected for a long time. While the small flame is not particularly hazardous if it remains small, the danger is that it might flare up if the air convection is suddenly increased or if the flame spreads into another fuel source.
Bibliography Type:	Description: (Last Updated: 12/19/2013)
Papers from Meeting Proceedings	Ferkul PV, Olson SL, Johnston MC, T'ien JS. "Flammability of a Thin Solid in Opposed and Concurrent Air Flow in Microgravity." 35th International Symposium on Combustion, San Francisco, CA, August 3-8, 2014. 35th International Symposium on Combustion, August 2014. Submitted in November 2013. , Nov-2013